

## A. Data → source, URL, description, variables+units, table

1. **Source name** (World Bank, World Development Indicators)
2. **Full URL** (<https://data.worldbank.org/indicator/NY.GDP.PCAP.CD?locations=PH>)
3. Description about the data represented:

The World Bank's World Development Indicators (WDI) database provided the GDP per capita (in current US dollars) for the Philippines, which served as the dataset for this lab. This indicator, which is expressed in current US dollars, calculates the average economic output per person in the nation during a specific year. Also, this data consists of annual observations from 2008 to 2022

4. x and y variables, with units (x = Year; y = GDP per capita in current US\$)

### **Variables:**

- Independent variable (x): Year (unit: years, e.g., 2008, 2009, ...)
- Dependent variable (y): GDP per capita, current US\$ (unit: US dollars per person)

5. Table of observations:

| Year (x) | GDP per capita, current US\$ (y) |
|----------|----------------------------------|
| 2008     | 1959.292                         |
| 2009     | 1864.45                          |
| 2010     | 2162.914                         |
| 2011     | 2383.919                         |
| 2012     | 2614.616                         |
| 2013     | 2781.280                         |
| 2014     | 2866.838                         |
| 2015     | 2909.858                         |
| 2016     | 2985.196                         |
| 2017     | 3038.121                         |
| 2018     | 3168.510                         |
| 2019     | 3400.789                         |
| 2020     | 3227.579                         |
| 2021     | 3484.386                         |
| 2022     | 3548.069                         |

## B. Python script → full code as text

```
# Lab 03 - Linear Regression by Least Squares
# Numerical Methods, Section 3D

import numpy as np
import matplotlib.pyplot as plt

# ----- 1. Data (replace with your verified values) -----
x = np.array([2008, 2009, 2010, 2011, 2012, 2013, 2014,
              2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022], dtype=float)
y = np.array([1925, 1851, 2155, 2364, 2598, 2762, 2873,
              2904, 2951, 3009, 3199, 3354, 3325, 3578, 3499], dtype=float)

n = len(x)

# ----- 2. Least squares formulas -----
#  $a_1 = (n \cdot \sum(xy) - \sum(x) \cdot \sum(y)) / (n \cdot \sum(x^2) - (\sum(x))^2)$ 
#  $a_0 = \text{mean}(y) - a_1 \cdot \text{mean}(x)$ 
sum_x = np.sum(x)
sum_y = np.sum(y)
sum_xy = np.sum(x * y)
sum_x2 = np.sum(x ** 2)

a1 = (n * sum_xy - sum_x * sum_y) / (n * sum_x2 - sum_x ** 2)
a0 = (sum_y / n) - a1 * (sum_x / n)

print(f"a0 (intercept) = {a0:.4f}")
print(f"a1 (slope) = {a1:.4f}")

# ----- 3. Fit statistics -----
y_hat = a0 + a1 * x # predicted values
Sr = np.sum((y - y_hat) ** 2) # SSE
St = np.sum((y - np.mean(y)) ** 2)
r2 = (St - Sr) / St
syx = np.sqrt(Sr / (n - 2)) # standard error of estimate

print(f"Sr (SSE) = {Sr:.4f}")
print(f"r^2 = {r2:.4f}")
print(f"s_y/x = {syx:.4f}")

# ----- 4. Prediction -----
x_new = 2025
y_new = a0 + a1 * x_new
```

```
print(f"Prediction for {x_new}: {y_new:.2f} US$")
```

```
# ----- 5. Graph: data + fitted line -----
```

```
plt.figure()
plt.scatter(x, y, label="Data")
plt.plot(x, y_hat, color="red", label=f"y = {a0:.1f} + {a1:.1f}x")
plt.xlabel("Year")
plt.ylabel("GDP per capita (current US$)")
plt.title("Philippines GDP per capita vs Year")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.savefig("fit.png", dpi=150)
```

```
# ----- 6. Residual plot -----
```

```
residuals = y - y_hat
plt.figure()
plt.axhline(0, color="black", linewidth=0.8)
plt.scatter(x, residuals)
plt.xlabel("Year")
plt.ylabel("Residual (y - y_hat)")
plt.title("Residual Plot")
plt.grid(True)
plt.tight_layout()
plt.savefig("residuals.png", dpi=150)
```

```
plt.show()
```

## C. Regression results

### THE FINAL SCRIPT IN TERMINAL:

```
PS C:\Users\Admin\OneDrive\Desktop\Numsol Exercises> & "C:\Program
Files\Python314\python.exe" "c:/Users/Admin/OneDrive/Desktop/Numsol
Exercises/Bernales_PatrickJohn_3D_Lab03.py"
a0 (intercept) = -231241.6398
a1 (slope)      = 116.1628
Sr (SSE)        = 222153.7324
r^2             = 0.9445
s_y/x           = 130.7239
Prediction for 2025: 3988.02 US$
```

### INTERPRETATION FOR THE SCRIPT:

**Slope ( $a_1 = 116.1628$ )**. According to the slope, the Philippines' GDP per capita grew by about 116.16 US dollars annually on average during the study period. Put differently, the model predicts that each Filipino's share of economic output will increase by roughly 116 US dollars for every extra year. The data's overall upward economic trend is supported by this positive slope.

**Intercept ( $a_0 = -231,241.6398$ )**. The value that the model would predict in Year 0, or when  $x = 0$ , is known as the intercept. The intercept has no actual economic significance in this case because Year 0 is well outside the data range (2008–2022). It merely serves as a constant that appropriately places the line through the actual data points; it does not imply that GDP per capita was negative in Year 0. Because of this, it is not appropriate to interpret the intercept alone in this situation.

**Standard error ( $s_{\{y/x\}} = 130.7239$  US)**: The estimate's standard error is roughly 130.72 u.s. This indicates that the average difference between the actual and line-predicted GDP per capita values is approximately  $\pm 131$  US. The conclusion that the linear model fits the data well is supported by the fact that the error of roughly 131 US is relatively small given that the GDP per capita values in the data set range from roughly 1,851 US to 3,578 US.

**Residuals**. The residual plot (residual =  $y - \hat{y}$ ) displays the discrepancy between each actual value and its predicted value. A straight line is a plausible model for this data, as the residuals seem to be dispersed fairly near the zero line without a strong systematic pattern. However, some points may be farther from the line than others because the values cover a time period that includes economic fluctuations (like the 2020 dip during the pandemic). Overall, the residuals do not exhibit a distinct curve or fanning pattern that would indicate the linear model is inappropriate, although these larger residuals show years when actual GDP per capita deviated more from the long-term linear trend.

Overall conclusion. The regression shows a clear, strong, and positive linear relationship between year and GDP per capita for the Philippines from 2008 to 2022. The high  $r^2$  value and the relatively small standard error indicate that the linear model describes the data well, and the positive slope reflects steady economic growth over the period



